Aarush Ram Anandh

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EXECUTIVE SUMMARY

Python/PyTorch researcher working on generative models, reinforcement learning and optimal transport.
Hands-on MaskablePPO/DQN implementation and multi-seed experiments with statistical evaluation.
Public diffusion preprint; RT-DETR rare-class correct detection improved from 40% to 80%.

EDUCATION

Ashoka University — B.Sc. (Hons.), Computer Science and Mathematics **Expected 2028**

RESEARCH EXPERIENCE

Constrained Image-Synthesis Lab, Ashoka University **Sep 2025–Present**
Research Fellow

- Improved latent-diffusion colour fidelity to $\Delta E_{00} < 1$, with approximately **4× better colour accuracy** than evaluated open-source baselines; results reported in the preprint below.
- Developing **Controlled Generalised Sliced Wasserstein (CGSW)**, an optimal-transport framework with control-conditioned probes, projections and slice aggregation; paper submitted to **JMLR**.
- Implementing **C++** methods and machine-verified **Lean** specifications/proofs for CGSW.

Koita Centre for Digital Health, Ashoka University **May 2026–Present**
Research Intern (part-time)

- Contributed to a wearable-data pipeline achieving approximately **70% sensitivity (recall)** for **asthma exacerbation detection** in study evaluation.
- Built HR/HRV smoothing and change-point workflows for **191 patients**, with patient-wise evaluation; implemented experimental frozen time-series foundation-model routes, not yet run in cohort evaluation.

INDUSTRY EXPERIENCE

Avartan Labs **May–Jul 2026**
Product Engineering / Machine Learning Intern

- Doubled rare-class correct detection from **40% to 80%** in an **18-class RT-DETR** sorting pipeline.
- Investigated focal loss, logit adjustment, class-specific thresholds and targeted fine-tuning for rare classes; evaluated detector failure modes through crop-based refinement.

SELECTED AI PROJECTS

Reinforcement Learning for Pokemon Battles **Ongoing**
Two-Person RL Coursework Project [GitHub]

- Co-developed a **Gymnasium/poke-env** environment with **763 observation features**, legality masks over 26 actions, potential-based reward shaping, and **MaskablePPO and masked DQN** agents.
- Evaluated PPO across **two team variants and five seeds each**, at 200,000 decision steps per run; analysed convergence, Wilson confidence intervals and exact permutation tests.
- Recorded **99.7–100% final win rates** against a fixed heuristic opponent across three opponent teams, over 1,020 battles per run; found no significant learning-speed difference between team variants.

Monthly Equity Return Prediction **Completed May 2026**
Team Coursework Project [GitHub]

- Collaborated with **Tanush Seal** on a CatBoost financial forecasting pipeline comparing **seven residual approaches**, including Transformers and HMMs; engineered temporal features and company-level error diagnostics.

PUBLICATION

Inference-Time Loss-Guided Colour Preservation in Diffusion Sampling (2026).
Public preprint: arXiv:2601.17259; colour-fidelity research described above.

TECHNICAL SKILLS

Programming: Python, C++, C, Bash, Lean. **Deep learning:** PyTorch, Diffusers, Transformers.
RL / Tools: Gymnasium, MaskablePPO, DQN, Git, Jupyter notebooks.
Foundations: Linear algebra, probability, statistics, optimal transport; controlled experiments and model evaluation.